70216008-朴俊范-第八次实验

Task1.

创建db且添加两个用户

```
mysql> CREATE DATABASE db9
    -> ;
Query OK, 1 row affected (0.02 sec)

mysql> USE db9;
Database changed
mysql> CREATE TABLE user(
    -> uid INT PRIMARY KEY,
    -> name VARCHAR(255) NOT NULL,
    -> money DECIMAL(10,2) NOT NULL
    -> );
Query OK, 0 rows affected (0.07 sec)
```

```
mysql> INSERT INTO user VALUES (1,'A',2000);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO user VALUES (2,'B',3000);
Query OK, 1 row affected (0.01 sec)
```

备份数据库

```
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C:\Users\버미>mysqldump -u root -p db9 > db9_backup.sql
Enter password: **********
```

删除该表

```
mysql> DROP TABLE user;
Query OK, 0 rows affected (0.03 sec)
```

恢复数据库

```
C:\Users\버미>mysql -u root -p db9 <db9_backup.sql
Enter password: **********
```

```
mysql> USE db9;
Database changed
mysql> SHOW TABLES;
+---------------+
| Tables_in_db9 |
+---------------+
| user          |
+---------------+
1 row in set (0.00 sec)

mysql> SELECT * FROM user;
+-----+------+---------+
| uid | name | money   |
+-----+------+---------+
|   1 | A    | 2000.00 |
|   2 | B    | 3000.00 |
+-----+------+---------+
2 rows in set (0.00 sec)

mysql>
```

Task2

创建表添加用户且删除用户

```
mysql> CREATE TABLE user2(
    -> uid INT PRIMARY KEY,
    -> name VARCHAR(50) NOT NULL,
    -> money DECIMAL(10,2) NOT NULL
    -> );
Query OK, 0 rows affected (0.03 sec)

mysql> INSERT INTO user2 VALUES (1,'A',2000);
Query OK, 1 row affected (0.01 sec)

mysql> INSERT INTO user2 VALUES (2,'B',3000);
Query OK, 1 row affected (0.01 sec)

mysql> DELETE FROM user2 WHERE name = 'A';
Query OK, 1 row affected (0.01 sec)

mysql> DELETE FROM user2 WHERE name = 'B';
Query OK, 1 row affected (0.01 sec)
```

查找位置（以下省略）

```
mysql> show binlog events in 'DESKTOP-1CB6VL8-bin.000039';
+-----------------------------+------+----------------+-----------+-------------+-----------------------------------
-----------------------------------------------------------------+
| Log_name                    | Pos  | Event_type     | Server_id | End_log_pos | Info
                                                                |
+-----------------------------+------+----------------+-----------+-------------+-----------------------------------
-----------------------------------------------------------------+
| DESKTOP-1CB6VL8-bin.000039 |    4 | Format_desc    |         1 |         126 | Server ver: 8.0.36, Binlog ver: 4
                                                                |
| DESKTOP-1CB6VL8-bin.000039 |  126 | Previous_gtids |         1 |         157 |
                                                                |
| DESKTOP-1CB6VL8-bin.000039 |  157 | Anonymous_Gtid |         1 |         234 | SET @@SESSION.GTID_NEXT= 'ANONYMOUS'
                                                                |
| DESKTOP-1CB6VL8-bin.000039 |  234 | Query          |         1 |         339 | CREATE DATABASE db9 /* xid=3 */
                                                                |
| DESKTOP-1CB6VL8-bin.000039 |  339 | Anonymous_Gtid |         1 |         418 | SET @@SESSION.GTID_NEXT= 'ANONYMOUS'
                                                                |
| DESKTOP-1CB6VL8-bin.000039 |  418 | Query          |         1 |         602 | use `db9`; CREATE TABLE user(
uid INT PRIMARY KEY,
name VARCHAR(255) NOT NULL,
money DECIMAL(10,2) NOT NULL
) /* xid=6 */                                                                                                          |
| DESKTOP-1CB6VL8-bin.000039 |  602 | Anonymous_Gtid |         1 |         681 | SET @@SESSION.GTID_NEXT= 'ANONYMOUS'
                                                                |
| DESKTOP-1CB6VL8-bin.000039 |  681 | Query          |         1 |         755 | BEGIN
                                                                |
| DESKTOP-1CB6VL8-bin.000039 |  755 | Table_map      |         1 |         815 | table_id: 85 (db9.user)
                                                                |
| DESKTOP-1CB6VL8-bin.000039 |  815 | Write_rows     |         1 |         863 | table_id: 85 flags: STMT_END_F
                                                                |
| DESKTOP-1CB6VL8-bin.000039 |  863 | Xid            |         1 |         894 | COMMIT /* xid=8 */
                                                                |
| DESKTOP-1CB6VL8-bin.000039 |  894 | Anonymous_Gtid |         1 |         973 | SET @@SESSION.GTID_NEXT= 'ANONYMOUS'
                                                                |
| DESKTOP-1CB6VL8-bin.000039 |  973 | Query          |         1 |        1047 | BEGIN
                                                                |
| DESKTOP-1CB6VL8-bin.000039 | 1047 | Table_map      |         1 |        1107 | table_id: 85 (db9.user)
                                                                |
| DESKTOP-1CB6VL8-bin.000039 | 1107 | Write_rows     |         1 |        1155 | table_id: 85 flags: STMT_END_F
                                                                |
| DESKTOP-1CB6VL8-bin.000039 | 1155 | Xid            |         1 |        1186 | COMMIT /* xid=9 */
```

使用位置恢复（以下省略）

```
C:₩Windows₩System32>mysqlbinlog --no-defaults --start-position=4397 --stop-position=4689 "C:₩ProgramData₩MySQL₩MySQL Server 8.0₩Data₩DESKTOP-1CB6VL8-bin.000039" | mysql -u root -p
Enter password: **********
# The proper term is pseudo_replica_mode, but we use this compatibility alias
# to make the statement usable on server versions 8.0.24 and older.
/*!50530 SET @@SESSION.PSEUDO_SLAVE_MODE=1*/;
/*!50003 SET @OLD_COMPLETION_TYPE=@@COMPLETION_TYPE,COMPLETION_TYPE=0*/;
DELIMITER /*!*/;
# at 157
#240522  8:03:31 server id 1  end_log_pos 126 CRC32 0x78d258fc  Start: binlog v 4, server v 8.0.36 created 240522  8:03:31 at startup
# Warning: this binlog is either in use or was not closed properly.
ROLLBACK/*!*/;
BINLOG '
UzZNZg8BAAAAegAAAH4AAAAEAAQAOC4wLjM2AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAABTNk1mEwANAAgAAAAABAAEAAAAYgAEGggAAAAICAgCAAAACgoKKioAEjQA
CigAAfxYOng=
'/*!*/;
# at 4397
#240523 17:48:58 server id 1  end_log_pos 4476 CRC32 0x3b1189c8  Anonymous_GTID  last_committed=16  sequence_number=17  rbr_only=yes  original_committed_timestamp
38204286  immediate_commit_timestamp=1716457738204286  transaction_length=292
/*!50718 SET TRANSACTION ISOLATION LEVEL READ COMMITTED*//*!*/;
# original_commit_timestamp=1716457738204286 (2024-05-23 17:48:58.204286 중국 표준시)
# immediate_commit_timestamp=1716457738204286 (2024-05-23 17:48:58.204286 중국 표준시)
/*!80001 SET @@session.original_commit_timestamp=1716457738204286*//*!*/;
/*!80014 SET @@session.original_server_version=80036*//*!*/;
/*!80014 SET @@session.immediate_server_version=80036*//*!*/;
SET @@SESSION.GTID_NEXT= 'ANONYMOUS'/*!*/;
# at 4476
#240523 17:48:58 server id 1  end_log_pos 4550 CRC32 0x77547788  Query  thread_id=17  exec_time=0  error_code=0
SET TIMESTAMP=1716457738/*!*/;
SET @@session.pseudo_thread_id=17/*!*/;
SET @@session.foreign_key_checks=1, @@session.sql_auto_is_null=0, @@session.unique_checks=1, @@session.autocommit=1/*!*/;
SET @@session.sql_mode=1168113696/*!*/;
SET @@session.auto_increment_increment=1, @@session.auto_increment_offset=1/*!*/;
/*!₩C euckr *//*!*/;
SET @@session.character_set_client=19,@@session.collation_connection=19,@@session.collation_server=255/*!*/;
SET @@session.lc_time_names=0/*!*/;
SET @@session.collation_database=DEFAULT/*!*/;
/*!80011 SET @@session.default_collation_for_utf8mb4=255*//*!*/;
BEGIN
/*!*/;
# at 4550
#240523 17:48:58 server id 1  end_log_pos 4611 CRC32 0x93d545a7  Table_map: `db9`.`user2` mapped to number 99
# has_generated_invisible_primary_key=0
# at 4611
#240523 17:48:58 server id 1  end_log_pos 4658 CRC32 0x46708f6d  Write_rows: table id 99 flags: STMT_END_F
```

查找时间（以下省略）

```
C:₩Windows₩System32>mysqlbinlog --no-defaults "C:₩ProgramData₩MySQL₩MySQL Server 8.0₩Data₩DESKTOP-1CB6VL8-bin.000039"
# The proper term is pseudo_replica_mode, but we use this compatibility alias
# to make the statement usable on server versions 8.0.24 and older.
/*!50530 SET @@SESSION.PSEUDO_SLAVE_MODE=1*/;
/*!50003 SET @OLD_COMPLETION_TYPE=@@COMPLETION_TYPE,COMPLETION_TYPE=0*/;
DELIMITER /*!*/;
# at 4
#240522  8:03:31 server id 1  end_log_pos 126 CRC32 0x78d258fc  Start: binlog v 4, server v 8.0.36 created 240522  8:03:31 at startup
# Warning: this binlog is either in use or was not closed properly.
ROLLBACK/*!*/;
BINLOG '
UzZNZg8BAAAAegAAAH4AAAABAAQAOC4wLjM2AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAAAABTNk1mEwANAAgAAAAABAAEAAAAYgAEGggAAAAICAgCAAAACgoKKioAEjQA
CigAAfxYOng=
'/*!*/;
# at 126
#240522  8:03:31 server id 1  end_log_pos 157 CRC32 0xe18dcead  Previous-GTIDs
# [empty]
# at 157
#240523 16:08:07 server id 1  end_log_pos 234 CRC32 0xa55a8ccb  Anonymous_GTID  last_committed=0       sequence_number=1       rbr_only=no
   immediate_commit_timestamp=1716451687554765
        transaction_length=182
# original_commit_timestamp=1716451687554765 (2024-05-23 16:08:07.554765 중국 표준시)
# immediate_commit_timestamp=1716451687554765 (2024-05-23 16:08:07.554765 중국 표준시)
/*!80001 SET @@session.original_commit_timestamp=1716451687554765*//*!*/;
/*!80014 SET @@session.original_server_version=80036*//*!*/;
/*!80014 SET @@session.immediate_server_version=80036*//*!*/;
SET @@SESSION.GTID_NEXT= 'ANONYMOUS'/*!*/;
# at 234
#240523 16:08:07 server id 1  end_log_pos 339 CRC32 0xab68fe93  Query   thread_id=8     exec_time=0     error_code=0
        Xid = 3
SET TIMESTAMP=1716451687/*!*/;
SET @@session.pseudo_thread_id=8/*!*/;
SET @@session.foreign_key_checks=1, @@session.sql_auto_is_null=0, @@session.unique_checks=1, @@session.autocommit=1/*!*/;
SET @@session.sql_mode=1168113696/*!*/;
SET @@session.auto_increment_increment=1, @@session.auto_increment_offset=1/*!*/;
/*!₩C euckr *//*!*/;
SET @@session.character_set_client=19,@@session.collation_connection=19,@@session.collation_server=255/*!*/;
SET @@session.lc_time_names=0/*!*/;
SET @@session.collation_database=DEFAULT/*!*/;
/*!80011 SET @@session.default_collation_for_utf8mb4=255*//*!*/;
/*!80016 SET @@session.default_table_encryption=0*//*!*/;
CREATE DATABASE db9
/*!*/;
# at 339
#240523 16:10:03 server id 1  end_log_pos 418 CRC32 0xe2755ea4  Anonymous_GTID  last_committed=1       sequence_number=2       rbr_only=no
```

使用时间恢复

```
:₩Windows₩System32>mysqlbinlog --no-defaults --start-datetime="2024-05-23 16:11:45" --stop-d
tetime="2024-05-23 16:17:22" "C:₩ProgramData₩MySQL₩MySQL Server 8.0₩Data₩DESKTOP-1CB6VL8-bin
000039" -u root -p
nter password: **********
 The proper term is pseudo_replica_mode, but we use this compatibility alias
 to make the statement usable on server versions 8.0.24 and older.
*!50530 SET @@SESSION.PSEUDO_SLAVE_MODE=1*/;
*!50003 SET @OLD_COMPLETION_TYPE=@@COMPLETION_TYPE,COMPLETION_TYPE=0*/;
ELIMITER /*!*/;
 at 4
240522  8:03:31 server id 1  end_log_pos 126 CRC32 0x78d258fc  Start: binlog v 4, server v 8
0.36 created 240522  8:03:31 at startup
 Warning: this binlog is either in use or was not closed properly.
OLLBACK/*!*/;
INLOG '
zZNZg8BAAAAegAAAH4AAAABAAQAOC4wLjM2AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
AAAAAAAAAAAAAAAABTNk1mEwANAAgAAAAABAAEAAAAYgAEGggAAAAICAgCAAAACgoKKioAEjQA
igAAfxYOng=
/*!*/;
 at 894
240523 16:11:45 server id 1  end_log_pos 973 CRC32 0x4b8ed7ec  Anonymous_GTID  last_committe
=3     sequence_number=4       rbr_only=yes    original_committed_timestamp=1716451905658063
       immediate_commit_timestamp=1716451905658063     transaction_length=292
*!50718 SET TRANSACTION ISOLATION LEVEL READ COMMITTED*//*!*/;
 original_commit_timestamp=1716451905658063 (2024-05-23 16:11:45.658063 중국 표준시)
 immediate_commit_timestamp=1716451905658063 (2024-05-23 16:11:45.658063 중국 표준시)
*!80001 SET @@session.original_commit_timestamp=1716451905658063*//*!*/;
*!80014 SET @@session.original_server_version=80036*//*!*/;
*!80014 SET @@session.immediate_server_version=80036*//*!*/;
ET @@SESSION.GTID_NEXT= 'ANONYMOUS'/*!*/;
 at 973
240523 16:11:45 server id 1  end_log_pos 1047 CRC32 0x0bbefcba         Query   thread_id=8
       exec_time=0     error_code=0
ET TIMESTAMP=1716451905/*!*/;
ET @@session.pseudo_thread_id=8/*!*/;
ET @@session.foreign_key_checks=1, @@session.sql_auto_is_null=0, @@session.unique_checks=1,
@session.autocommit=1/*!*/;
ET @@session.sql_mode=1168113696/*!*/;
ET @@session.auto_increment_increment=1, @@session.auto_increment_offset=1/*!*/;
*!₩C euckr *//*!*/;
ET @@session.character_set_client=19,@@session.collation_connection=19,@@session.collation_s
rver=255/*!*/;
ET @@session.lc_time_names=0/*!*/;
ET @@session.collation_database=DEFAULT/*!*/;
*!80011 SET @@session.default_collation_for_utf8mb4=255*//*!*/;
EGIN
*!*/;
 at 1047
240523 16:11:45 server id 1  end_log_pos 1107 CRC32 0x44b8ffa5         Table_map: `db9` `use
```

```
mysql> select * from user2;
+-----+------+---------+
| uid | name | money   |
+-----+------+---------+
|   1 | A    | 2000.00 |
|   2 | B    | 3000.00 |
+-----+------+---------+
2 rows in set (0.00 sec)

mysql>
```